DUYGU KAYTAN

Robotics & Autonomous Systems Engineer

Cyber-Physical Systems | Control | AI | Embedded Systems

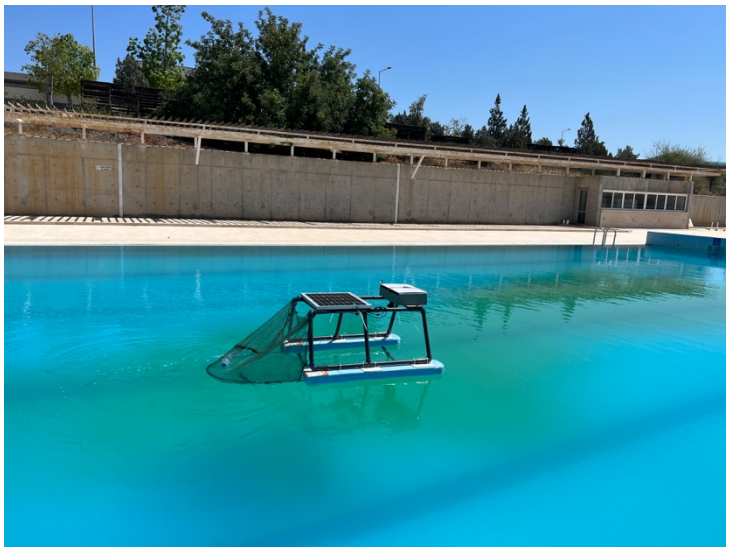
PROFILE

Master's student in Intelligent Transportation Systems with strong hands-on experience in robotics, control systems, and embedded systems. Focused on autonomous platforms, sensor integration, and real-world system deployment. Experienced in designing and implementing end-to-end robotic systems, combining hardware and software to build intelligent cyber-physical systems. Demonstrated ability to develop, test, and validate engineering solutions in real-world environments. Actively conducting research in autonomous systems, with multiple publications and submissions in leading journals and conferences.

PROJECTS

Autonomous Solar-Powered Surface Cleaning Robot

Multidisciplinary Engineering Project (ESC 491–492), Middle East Technical University (METU)

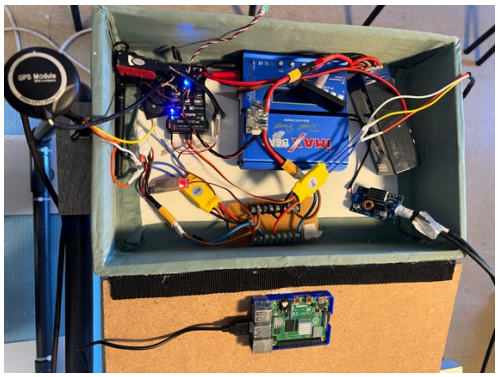


Designed and implemented a solar-powered autonomous surface vehicle for collecting floating waste, integrating mechanical design, embedded control, and energy systems within a multidisciplinary engineering project.

Developed in collaboration with a computer engineering team, focusing on full system integration, real-world deployment, and performance validation.

Key Contributions

- Designed and implemented the complete hardware architecture, including power distribution, motor systems, and physical platform
- Developed a solar-powered energy system with battery management and efficient power utilization
- Integrated GPS, telemetry, and remote control systems for real-time operation
- Led system integration, field testing, and validation in real-world conditions
- Collaborated on control software and mission planning with a multidisciplinary team



Demo Video: https://www.youtube.com/watch?v=cjh0bYrHCVc

Experimental Robotics & Embedded Systems Development

Independent Research & Personal Laboratory Work

Conducted hands-on development and experimental research on autonomous robotic systems in a self-built laboratory environment, focusing on real-world system integration, robustness, and intelligent behavior under uncertainty.

Worked on multiple robotic platforms including mobile rovers, small-scale assistant robots, unmanned surface vehicles, and UAV systems, combining hardware and software to develop cyber-physical systems.

Key Areas of Work

- Designed and developed mobile robotic platforms (rovers, boats, and small autonomous systems)
- Integrated multi-sensor setups including LiDAR, IMU, GPS, encoder, and ToF sensors
- Performed state estimation and sensor fusion for real-time system awareness
- Developed anomaly detection approaches for identifying faulty sensor behavior
- Explored robust system design and safe navigation strategies under uncertainty
- Conducted circuit design, soldering, embedded system debugging, and hardware integration
- Built and tested hybrid systems combining control, AI, and real-world sensing

This hands-on experimental work directly supports my ongoing research and publication activities in autonomous systems and intelligent control.

Teknofest 2022 – Combat UAV Project (Fighting UAV Category)

Multidisciplinary Engineering Team, Middle East Technical University

Contributed to the design and development of an autonomous UAV system for aerial combat scenarios within a multidisciplinary engineering team.

The system integrates autonomous flight control, computer vision-based target detection, real-time telemetry, and ground control station interaction.

Key Contributions

- Participated in system-level design of an autonomous UAV platform
- Contributed to integration of Pixhawk-based flight control and Raspberry Pi-based onboard computation
- Supported development of communication architecture including telemetry and ground control interaction
- Assisted in system integration, testing workflows, and validation procedures
- Collaborated in a multidisciplinary team covering hardware, software, and control subsystems
- Successfully completed all technical design and evaluation phases of the Teknofest UAV competition

Research & Publications

My research focuses on state estimation, anomaly detection, and safety-oriented decision-making in autonomous systems.

Akıllı Tahminciler: Otonom Sürüş için Pekiştirmeli Öğrenme ile Optimize Edilmiş Unscented Kalman Filtresi

26. Otomatik Kontrol Ulusal Konferansı (TOK 2025) – Published

- Proposed a reinforcement learning-enhanced UKF for autonomous vehicle state estimation
- Developed and validated in MATLAB/Simulink environment
- Improved estimation performance under nonlinear and noisy conditions
- Demonstrated robustness across multiple driving scenarios

Açıklanabilir IMU Anomali Tespiti: Gürültü Enjeksiyonu Altında Makine Öğrenmesi Performansına Pencere Boyutunun Etkisi

SIU 2026 – Under Review

- Investigates IMU anomaly detection under controlled noise injection
- Compares multiple ML models for anomaly classification
- Uses SHAP for interpretability
- Evaluates impact of temporal window size on detection performance

Adaptive and Interpretable Reinforcement Learning-Enhanced Unscented Kalman Filtering for Autonomous Driving State Estimation

Applied Soft Computing – Major Revision Submitted

- Proposes interpretable RL-based adaptive covariance tuning for UKF
- Maintains physical interpretability while improving estimation accuracy
- Introduces structured interaction between RL agent and filtering system
- Validated with statistical significance analysis across multiple scenarios

A Safety-Oriented Optimization Approach for Autonomous Navigation

Robotics and Autonomous Systems – Under Review

- Introduces a safety-aware extension to optimization-based navigation
- Implements inhibitory mechanism to suppress unsafe trajectories
- Improves robustness in dynamic and uncertain environments
- Designed for real-world robotic navigation systems

Ongoing Research: Explainable Encoder-Based Anomaly Detection in Mobile Robots

- Developing anomaly detection system using real-world encoder data
- Injecting controlled faults for robustness evaluation
- Comparing ML models for classification performance
- Integrating explainable AI for transparent decision-making
- Targeting submission to robotics-focused journals
- Designing a real-time human–machine interface for system monitoring, anomaly visualization, and human-in-the-loop decision support

MSc Thesis (Ongoing)

Real-time Anomaly Detection and Response Mechanism Against Sensor Spoofing Attacks in Autonomous Ground Vehicles Using Explainable AI

- Developing a real-time anomaly detection and response framework for autonomous ground vehicles
- Investigating sensor spoofing attacks across multi-sensor systems (LiDAR, IMU, encoder, and other onboard sensors)
- Designing robust detection mechanisms using machine learning and explainable AI techniques
- Integrating anomaly-aware decision-making into autonomous navigation for improved system safety
- Exploring adaptive strategies to determine sensor reliability under adversarial or faulty conditions
- Implementing a human-in-the-loop monitoring interface for real-time system visualization and control

Technical Skills

Programming & Simulation

- Python (data processing, machine learning, simulation)
- MATLAB & Simulink (control systems, state estimation, system modeling)
- C/C++ (embedded systems and low-level control)

Robotics & Autonomous Systems

- Autonomous system design and integration
- State estimation (Kalman Filter, Unscented Kalman Filter)
- Sensor fusion (LiDAR, IMU, encoder, GPS)
- Real-time system implementation

AI & Data-Driven Methods

- Machine learning for anomaly detection
- Reinforcement learning for adaptive systems
- Explainable AI (model interpretability and decision analysis)

Embedded & Hardware Systems

- Embedded systems (ESP32, Teensy 4.1, Raspberry Pi, microcontroller-based systems)
- Sensor integration and real-world data acquisition
- Power systems and energy-aware design (including solar-based systems)

Tools & Frameworks

- ROS (Robot Operating System)
- Git & GitHub
- Simulation-driven development (MATLAB/Simulink environments)

Engineering Focus Areas

- Cyber-Physical Systems
- Autonomous Navigation & Safety
- Robust System Design
- Human-in-the-loop Systems